CSE 714 – AI Lab

CROPDX

Crop Disease Diagnosis Expert System

A rule-based AI system in Turbo Prolog for agricultural disease identification

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Presentation Overview

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Problem Statement

Why crop disease diagnosis matters in rural Bangladesh

02

System Design

Architecture, flowchart, and knowledge base design

03

Knowledge Base

4 crops, 8 diseases, 24 diagnostic rules

04

Prolog Implementation

Key predicates, inference engine, cut operator

05

Results & Demo

Sample session: Rice Blast diagnosis

06

Advantages & Limitations

Strengths, constraints, and future roadmap

Problem Statement

30%

annual yield loss
from Rice Blast alone

5 Core Pain Points

→ Geographical isolation from experts

→ Symptom misidentification

→ Delayed treatment & spread

→ No internet in rural areas

→ Pesticide overuse & resistance

The Agricultural Crisis

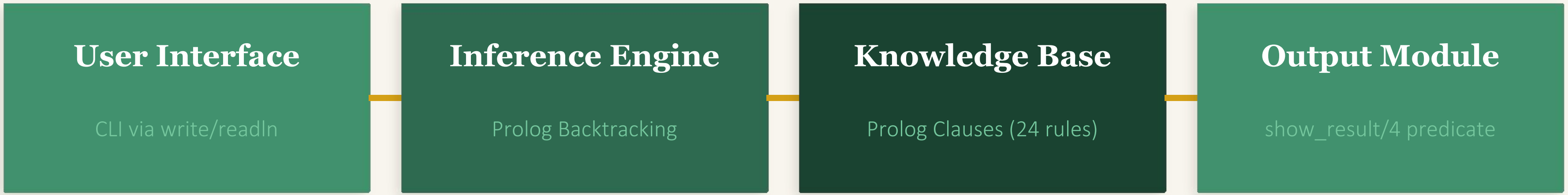
Crop diseases are a leading cause of yield loss in developing economies. In Bangladesh, farmers in rural areas have minimal access to plant pathologists — expert diagnosis can take days or weeks, allowing diseases to spread and devastate harvests.

Proposed Solution

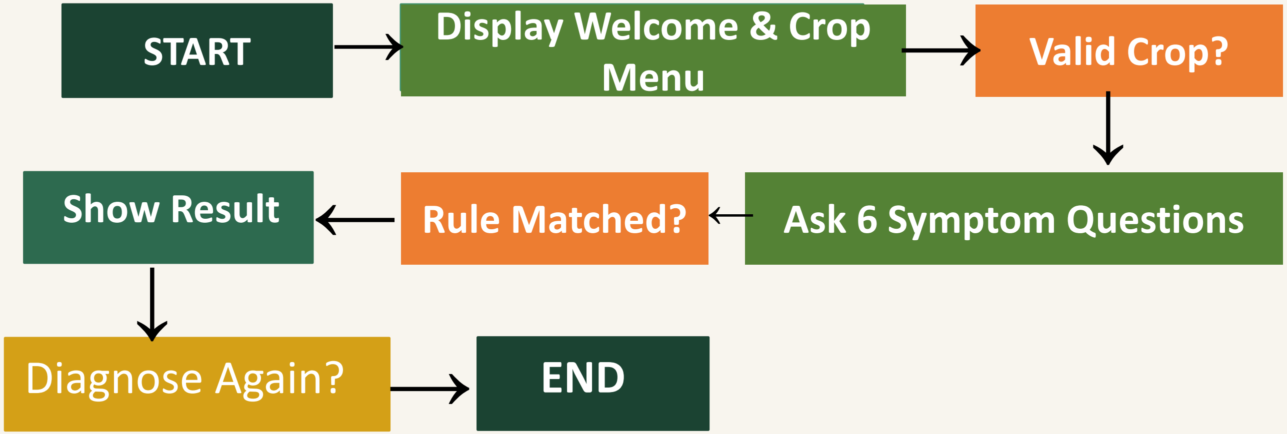
CROPDX — An offline, rule-based expert system in Turbo Prolog that:

- Runs on minimal hardware (DOSBox on any PC)
- Requires no internet connection
- Provides instant diagnosis in under 2 minutes
- Gives actionable treatment + prevention advice
- Is extensible without modifying the inference engine

System Architecture & Design



System Flowchart



Tools & Environment

Component	Details
Language	Turbo Prolog
Platform	DOSBox 0.74 on Windows
Compilation	Alt+C (internal compiler)
Execution	F9 (compile & run)
Source File	CROPDX.PRO

Knowledge Base — 4 Crops, 8 Diseases

RICE

Disease 1

Rice Blast

Magnaporthe oryzae

Diamond spots, gray center, brown border

Disease 2

Bacterial Leaf Blight

Xanthomonas oryzae

Yellow tips, water-soaked streaks, tip drying

POTATO

Disease 1

Late Blight

Phytophthora infestans

Dark water-soaked patches, white mold, tuber rot

Disease 2

Early Blight

Alternaria solani

Small brown spots, yellow halo, lower leaves first

TOMATO

Disease 1

Mosaic Virus

Tomato Mosaic Virus

Mosaic pattern, leaf curl, stunted growth

Disease 2

Fusarium Wilt

Fusarium oxysporum

Lower wilting, night recovery, brown stem

WHEAT

Disease 1

Yellow Rust

Puccinia striiformis

Yellow pustules in rows, pale yellowing

Disease 2

Powdery Mildew

Blumeria graminis

White powdery patches, rubs off, upper leaf

Turbo Prolog Implementation

Program Structure

`domains`	Type declarations (name = string, answer = string)
`predicates`	Predicate signatures for all 23 predicates
`clauses`	Diagnostic rules, menus, I/O, session logic
`goal`	Entry point → start.

Key Predicates

`diagnose_[crop]/6`	Core diagnostic engine — pattern matches 6 symptom answers
`ask_question/2`	Prompts user, captures yes/no via readln/1
`show_result/4`	Formats Disease, Cause, Treatment, Prevention
`show_unknown/0`	Graceful fallback when no pattern matches
`check_again/1`	Session loop — restart or exit after diagnosis

Key Insight: The cut operator (!) commits to the first matching clause, preventing incorrect backtracking to the wildcard show_unknown clause.

Sample Diagnosis — Rice Blast

```
• • •   CROPDX.PRO — Turbo Prolog

CROP DISEASE DIAGNOSIS EXPERT SYSTEM
AI Lab Project | Turbo Prolog 2.0

Select your crop:
rice / potato / tomato / wheat / exit
Enter crop name: rice

Crop selected: RICE
Answer each question with yes or no:

>> Diamond-shaped spots on leaves?  yes
>> Spots gray/white in center?      yes
>> Spots have brown/dark borders?   yes
>> Leaf tips turning yellow?        no
>> Water-soaked streaks on margins? no
>> Leaves drying from tips down?    no
```

DIAGNOSIS RESULT

DISEASE

Rice Blast

CAUSE

Magnaporthe oryzae (fungus)

TREATMENT

Apply Tricyclazole or Carbendazim. Remove infected parts.

PREVENTION

Use resistant varieties. Avoid excess nitrogen. Proper spacing.

Advantages & Limitations

✓ Advantages

Offline Operation
No internet required — any DOS-capable hardware

Transparent Reasoning
Readable Prolog rules, auditable by experts

Extensible
Add crops/diseases by appending clauses only

Consistent Advice
Uniform, unbiased recommendations every time

Low Cost
DOSBox (free) + standard PC, no cloud subscription

Educational Value
Demonstrates knowledge representation & backtracking

Session-Aware
Multiple diagnoses per run without restart

✗ Limitations

Binary Symptoms Only
Only yes/no — no severity levels or measurements

Limited Coverage
Only 2 diseases per crop; many real diseases unhandled

No Image Input
Cannot process photographs or sensor data

No Certainty Factors
No probability weighting for partial symptom matches

CLI Interface
Text-based; challenging for semi-literate users

Static Knowledge Base
Cannot learn from new cases; manual update needed

No Multi-Disease
Cannot diagnose two simultaneous diseases per session

Future Improvements

1 Expand Knowledge Base

Cover 20+ diseases per crop using phytopathology databases and community-sourced data

4 Multilingual Support

Bengali, Hindi, and other regional languages for rural farmer accessibility

2 Certainty Factors

Assign Mycin-style confidence weights (CF values) for partial symptom matches

5 Mobile Application

Android app with offline SQLite knowledge base for in-field diagnosis

3 Image-Based Input

Integrate CNN leaf classifier (PlantVillage model) to replace manual symptom entry

6 ML Integration

Hybrid system combining rule-based diagnosis with ML-trained classifiers

Roadmap Vision

Turbo Prolog CLI (Current)

▼

SWI-Prolog + Web UI

▼

Mobile (Android)

▼

Hybrid ML + Rules

▼

National Advisory Network

Conclusion

CROPDX successfully demonstrates a rule-based expert system for crop disease diagnosis using Turbo Prolog — correctly identifying 8 diseases across 4 staple crops.

- ✓ Multi-crop coverage (Rice, Potato, Tomato, Wheat)
- ✓ Dual-disease differentiation per crop
- ✓ Structured output: Disease → Cause → Treatment → Prevention
- ✓ Graceful fallback for unrecognised symptom patterns
- ✓ Session management for multiple diagnoses per run
- ✓ Full Turbo Prolog + DOSBox compatibility

Classical AI paradigms — expert systems, logic programming — retain significant practical value for offline, resource-constrained agricultural advisory applications.

Project Info

Student	**Humayra Adiba**
ID	**22701050**
Course	**CSE 714 – AI Lab**
Language	**Turbo Prolog**
Crops	**4**
Diseases	**8**
Rules	**24**
Date	**April 30, 2026**

THANK YOU

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